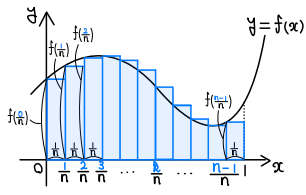


$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx$$

(ざっくり証明)

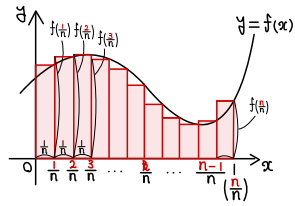


$$f\left(\frac{0}{n}\right) \cdot \frac{1}{n} + f\left(\frac{1}{n}\right) \cdot \frac{1}{n} + f\left(\frac{2}{n}\right) \cdot \frac{1}{n} + \cdots + f\left(\frac{n-1}{n}\right) \cdot \frac{1}{n}$$

$$\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) \cdot \frac{1}{n}$$

 $n \rightarrow \infty$ とすると

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx$$



$$f\left(\frac{1}{n}\right) \cdot \frac{1}{n} + f\left(\frac{2}{n}\right) \cdot \frac{1}{n} + f\left(\frac{3}{n}\right) \cdot \frac{1}{n} + \cdots + f\left(\frac{n}{n}\right) \cdot \frac{1}{n}$$

$$\sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \frac{1}{n}$$

 $n \rightarrow \infty$ とすると

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx$$

(例) $\lim_{n \rightarrow \infty} \frac{1}{n^3} (1^2 + 2^2 + 3^2 + \cdots + n^2) \cdots (*)$ について、次の各問いに答えよ。(1) $(*)$ の値を求めよ。(2) $(*)$ を定積分で表し、その値を求めよ。

$$\begin{aligned} (1) \quad (*) &= \lim_{n \rightarrow \infty} \frac{1}{n^3} \cdot \frac{1}{6} n(n+1)(2n+1) \\ &= \lim_{n \rightarrow \infty} \frac{1}{6} \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right) \\ &= \frac{1}{6} \cdot 1 \cdot 2 \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} (2) \quad (*) &= \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k^2 \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^2 \quad \text{長方形の面積の和} \\ &= \int_0^1 x^2 dx \\ &= \left[\frac{1}{3} x^3\right]_0^1 \\ &= \frac{1}{3} \end{aligned}$$

